

3.7: Joining Tables of Data

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Tables: address, country, city, customer, and payment

1- 10 countries for Rockbuster in terms of customer numbers:

Query		Query History
1	SELECT	D.country,
2	COUNT(customer_id) AS	count_cst_id
3	FROM	customer A
4	INNER JOIN	address B ON A.address_id = B.address_id
5	INNER JOIN	city C ON B.city_id = C.city_id
6	INNER JOIN	country D ON C.country_id = D.country_id
7	GROUP BY	country
8	ORDER BY	count_cst_id DESC
9	LIMIT	10;
10		

Data Output			Messages	Notifications
			SQL	Showing rows: 1 to 1
	country	count_cst_id		
	character varying (50)	bigint		
1	India	60		
2	China	53		
3	United States	36		
4	Japan	31		
5	Mexico	30		
6	Brazil	28		
7	Russian Federation	28		
8	Philippines	20		
9	Turkey	15		
10	Indonesia	14		

```
SELECT D.country,
COUNT(customer_id) AS count_cst_id
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
GROUP BY country
ORDER BY count_cst_id DESC
LIMIT 10;
```

a. Output:

Country	Count_cst_id
India	60
China	53
United States	36
Japan	31
Mexico	30
Brazil	28
Russian Federation	28
Philippines	20
Turkey	15
Indonesia	14

b. To write this query I started by connecting common columns in the tables, and how they would connect to each other. From that, I noticed that I would need 4 tables to get from the customer count to the country column. As every customer has a customer_id, I used this key factor as the source for my total customer per country. I also grouped by countries in order to have the total customers per each country. Since customer_id was present on customer table, and country was on country table, I had to use multiple joins to link them (customer table -> address table -> city table -> country table). For a more efficient query, I opted for the inner join, as I just needed limited data regarding customers count and countries. I also limited my data to 10, using a descending order, as I'm looking for the top 10 countries by customer volume.

2- Next, write a query to identify the top 10 cities that fall within the top 10 countries you identified in step 1. (Hint: the top 10 cities can be in any of the countries identified—you don't need to create a separate list for each country.)

Query Query History

```

1 SELECT C.city,
2 D.country,
3 COUNT(customer_id) AS count_cst_id
4 FROM customer A
5 INNER JOIN address B ON A.address_id = B.address_id
6 INNER JOIN city C ON B.city_id = C.city_id
7 INNER JOIN country D ON C.country_id = D.country_id
8 WHERE country IN ('India', 'China', 'United States', 'Japan', 'Mexico',
9 'Brazil', 'Russian Federation', 'Philippines', 'Turkey', 'Indonesia')
10 GROUP BY C.city, D.country
11 ORDER BY count_cst_id DESC
12 LIMIT 10;
13

```

Data Output Messages Notifications

city	country	count_cst_id
Aurora	United States	2
Atlixco	Mexico	1
Xintai	China	1
Adoni	India	1
Dhule (Dhule)	India	1
Kurashiki	Japan	1
Pingxiang	China	1
Sivas	Turkey	1
Ocala	Mexico	1
So Leopoldo	Brazil	1

```

SELECT C.city,
D.country,
COUNT(customer_id) AS count_cst_id
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE country IN ('India', 'China', 'United States', 'Japan',
'Mexico', 'Brazil', 'Russian Federation', 'Philippines',
'Turkey', 'Indonesia')
GROUP BY C.city, D.country
ORDER BY count_cst_id DESC
LIMIT 10;

```

To get to the 10 top cities within the 10 top countries, first I had to find out the 10 top countries, which steps were already explained on the previous question. From there, I used a new query, to retrieve limited data through multiple inner joins, so I could connect customer_id from customer table, city from city table, and country from country table.

This time, I grouped by cities and also ordered by cities in descending order, so I could get to the top cities with most customer_id count. By using the WHERE function, I was able to filter cities that were only located on the top 10 countries already outlined.

3- Top 5 customers from the top 10 cities who've paid the highest total amounts to Rockbuster:

Query	Query History	Scratch Pad
1	SELECT A.customer_id,	
2	A.first_name,	
3	A.last_name,	
4	C.city,	
5	D.country,	
6	SUM(amount) AS sum_payment_amount	
7	FROM payment E	
8	INNER JOIN customer A ON E.customer_id = A.customer_id	
9	INNER JOIN address B ON A.address_id = B.address_id	
10	INNER JOIN city C ON B.city_id = C.city_id	
11	INNER JOIN country D ON C.country_id = D.country_id	
12	WHERE city IN ('Aurora', 'Atlixco', 'Xintai', 'Adoni', 'Dhule (Dhulia)',	
13	'Kurashiki', 'Pingxiang', 'Sivas', 'Celaya', 'So Leopoldo')	
14	GROUP BY C.city, A.customer_id, A.first_name, A.last_name, D.country	
15	ORDER BY sum_payment_amount DESC	
16	LIMIT 5;	
17		

Data Output	Messages	Notifications
Showing rows: 1 to 5	Page No: 1	of 1
customer_id	first_name	last_name
integer	character varying (45)	character varying (45)
city	country	sum_payment_amount
character varying (30)	character varying (50)	numeric
84	Sara	Perry
518	Gabriel	Harder
587	Sergio	Stanfield
537	Clinton	Buford
367	Adam	Gooch

```

SELECT A.customer_id,
A.first_name,
A.last_name,
C.city,
D.country,
SUM(amount) AS sum_payment_amount
FROM payment E
INNER JOIN customer A ON E.customer_id = A.customer_id
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE city IN ('Aurora', 'Atlixco', 'Xintai', 'Adoni', 'Dhule (Dhulia)',
'Kurashiki', 'Pingxiang', 'Sivas', 'Celaya', 'So Leopoldo')
GROUP BY C.city, A.customer_id, A.first_name, A.last_name,
D.country

```

Output:

customer_id	first_name	last_name	city	country	sum_payment_amount
84	"Sara"	"Perry"	"Atlixco"	"Mexico"	128.70
518	"Gabriel"	"Harder"	"Sivas"	"Turkey"	108.75
587	"Sergio"	"Stanfield"	"Celaya"	"Mexico"	102.76
537	"Clinton"	"Buford"	"Aurora"	"United States"	98.76
367	"Adam"	"Gooch"	"Adoni"	"India"	97.80